Student Name	**ANSWERS**				
Teacher	AHU	COZ	GSL	HAL	MDI

CHEMISTRY
Unit 4, SAC 2
Covering concepts from Chapters 15.1 – 15.6 & 16 of Heinemann Chemistry 2
Food Chemistry

Week 10, Friday 17th September

Writing time: 45 minutes

Structure of book

Number of questions	Number of questions to be answered	Number of marks
5	5	27

- Students are permitted to bring into the assessment session:
  - Laptop computers, Data Book (electronic or printed)

Materials supplied
- Question and answer book of 7 pages by assignments on Teams.
- Data Book

Instructions
- Write your **name** and highlight your **teacher code** in the space provided above on this page.
- All written responses must be in English.
- Answer all questions in the spaces provided.

At the end of the assessment
- Ensure you have saved your work in Word and "Turned in" your task on Microsoft teams.
- Do not close the Word document until confirming that the file submitted into Teams has saved properly.

Students agree that all work completed in this SAC is their own work and they have not been assisted in the task by another individual.

Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

Instructions

Answer **all** questions in the spaces provided.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s)
- Chemical formulas can be written using regular size numbers (e.g. ethanol will be CH3CH2OH, calcium ion will be Ca2+)

Question 1 (4 marks)

Amino acids are the monomers for proteins.

a. Explain why they are classified as alpha (α) amino acids, with reference to structure and bonding.

Alpha (α) amino acids must have a **central/second Carbon atom** (α-carbon) that is bonded to:

- an **amino** ($-NH_2$) group and a **carboxyl** (-COOH) group (1 mark)
- a unique side chain (R-group) and H atom

1 mark

b. Explain what distinguishes an amino acid as essential and non-essential.

An essential amino acid is one that cannot be synthesised by humans and must consume in diet, whereas a non-essential amino acid can be synthesised.

- Must mention ability to **synthesise** and consumption in **diet** (1 mark)

1 mark

c. Explain the protonation state of all acidic and basic groups in the amino acid, leucine at pH 11.

There is a greater concentration of hydroxide (OH^-) ions in a basic/alkaline solution. The excess OH^- accepts protons from both the amino and carboxyl groups of the zwitterion to form $-NH_2$ and $-COO^-$, respectively.

- Explain amino and **carboxyl groups are deprotonated** by excess base OH^- (1 mark)
- The amino acid will be in the **anionic form** and have a **net negative charge**. (1 mark)

2 marks

Question 2 (7 marks)

Enzymes are crucial for the reactions involved in the metabolism of food in the human body.

Salivary amylase has an optimum pH of 6.8, whereas in stomach pepsin has an optimum pH of 1.5.

a. Explain why different enzymes (amylase and pepsin) are required to catalyse different functions in the body.

The 3D **shape of the active site** for amylase and pepsin **are different**. Hence, they will catalyse different reactions due to **enzyme specificity** for a substrate (one optical isomer)

- specificity of enzymes – **active site** can only fit a particular substrate (1 mark)

1 mark

b. Describe what happens to the amylase produced in the mouth once it reaches the stomach.

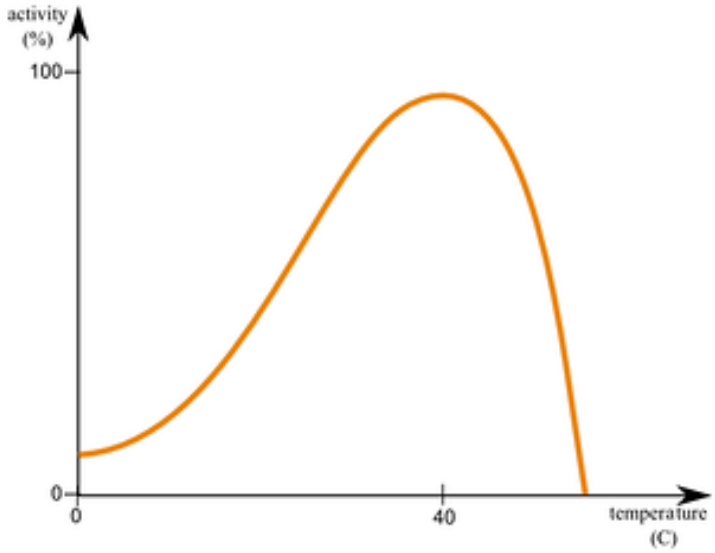
The low pH will affect the protonation states of the acidic/basic R-groups of amylase, thus the active site changes shape so it is no longer complementary to the substrate molecules.

Amylase operates effectively only within a narrow pH range around 6.8. The stomach pH of 1.5 is too low and will **denature amylase by disrupting the tertiary structure** (dipole-dipole/ionic bonds), therefore results in the **loss of catalytic activity**

- **Changes in pH environment** (1 mark)
- Denaturation of amylase – loss of 3ry structure **or** loss of active site shape and catalytic activity (1 mark)

2 marks

c. The graph below shows the rate of enzyme activity of pepsin in relation to temperature.



Use the graph to explain what happens to the enzyme activity of pepsin:

Note. Student **must** refer to graph and include optimal temperature (40 °C) of pepsin in responses

i. **below** the **optimal** temperature

At temperatures **lower than 40 °C**, particles have **lower average kinetic energy** to collide/less energy to overcome activation energy.

This results in **decreased frequency of successful collisions** between pepsin and its substrates to form enzyme-substrate complexes, thus **low catalytic activity** of the enzyme.

- **Lower catalytic activity below 40°C** (1 mark)
- Decreased kinetic **energy** (or heat) results in decreased **frequency of successful collisions or** less **energy to overcome Ea** (1 mark)

2 marks

ii. **above** the **optimal** temperature

At temperature higher than 40 °C, particles have higher average kinetic energy which disrupts the intermolecular forces (H bonding, ionic bonding, disulphide links etc.) responsible for the secondary/tertiary/quaternary structure of pepsin.

This causes denaturation of pepsin as the active site is permanently changed, which prevents the enzyme-substrate complexes from being formed.

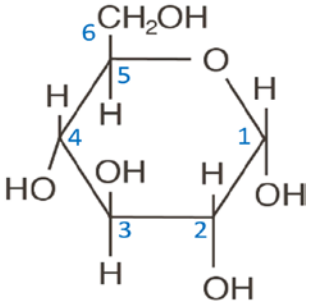
- **Lower catalytic activity/loss of activity above 40** (1 mark)
- Increased **heat/kinetic energy** will result in **denaturation**, disrupting interactions responsible for the tertiary structure. Change in the shape of the enzyme/active site and loss of activity or binding of substrate (1 mark)

2 marks

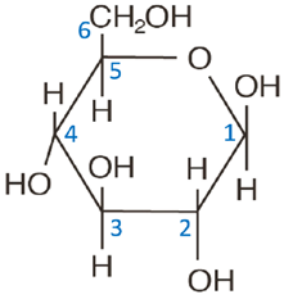
Question 3 (5 marks)

Glucose is one of the body's preferred sources of fuel which is obtained from bread, fruits, vegetables, and dairy products and it exists in two forms.

a. Explain whether the two different glucose molecules shown below are examples of stereoisomers.



Glucose molecule **I**



Glucose molecule **II**

Both glucose molecules (α-glucose and β-glucose) are stereoisomers because they have the same composition/chemical formula, but they differ in the 3-dimensional/spatial configuration of atoms/groups at one or more positions
In α-glucose, the OH group on the far right is down, and, in β-glucose, the OH group on the far right is up.

- They are stereoisomers as they have **same composition/chemical formula/atom connectivity/only differ in the 3D configuration** (1 mark).
- The difference is in the position of the **–OH in Carbon 1**. (1 mark)

2 marks

b. The following table lists a student's summary notes about glucose. It contains some correct and incorrect statements.

Number	Statement
1	Sucrose is a disaccharide formed by a condensation reaction between the monomers, glucose and fructose.
2	The monosaccharides glucose, galactose and fructose are structural isomers and they are highly soluble in water.
3	The monosaccharides glucose and galactose both have four chiral centres, but fructose only has three chiral centres.
4	Glucose molecules can polymerise to form two different forms of starch: amylopectin and amylose. Amylose is more soluble than amylopectin.

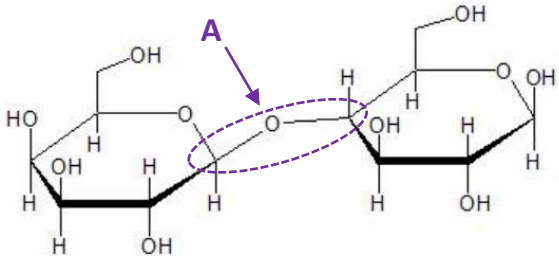
Identify **one incorrect statement** by writing the statement number in the table provided below. Explain why the statement is incorrect.

Statement number	Explanation
4	Yes, it is true that there are two different forms of starch formed from the glucose monosaccharide called amylose and amylopectin and that they have different solubilities. Amylose is a long linear polymer of starch formed by a-glucose molecules forming a glycosidic link between each glucose molecule. These long molecules coil into a spiral-helix and can pack very tightly together preventing the -OH groups in amylose to be in contact with the water to form H-bonding. Therefore, amylose is insoluble in cold water. Amylopectin is also formed from a-glucose molecules, but the glycosidic links are formed at different positions around the glucose rings which give rise to branches along the polysaccharide. This branching restricts the coiling of amylopectin, leaving more -OH groups exposed/in contact with water to allow it to dissolve as there is more H-bonding acting allowing it to dissolve • Correct statement (1 mark) • Compare structure of Amylose (**linear**) **and** Amylopectin (**branched**) and **ability to pack** (1 mark) • Explain solubility with reference to **H-bonding with water** (1 mark)
3	• Statement (1 mark) • Glucose and galactose have **five** chiral centres (1 mark) • Fructose has **four** chiral centres (1 mark)

3 marks

Question 4 (4 marks)

Lactose is a disaccharide which is naturally found in milk and milk products. It is typically broken down by lactase, an enzyme in the small intestine. Some people cannot digest lactose and are known to be lactose intolerant. The structure of lactose is shown below.



a. Name the link labelled as **A** in the lactose molecule shown above. Explain how this link is formed and name the type of reaction.

- **ether bond** or **glycosidic link** (1 mark)
- **Two hydroxyl/OH groups, one from each monosaccharide molecule, react/come together and release a water** molecule (1 mark)
- **Condensation** reaction (1 mark)

3 marks

b. Name the products formed during the hydrolysis of lactose.

- α-glucose/ B-glucose / glucose
- galactose

Must mention **both** monosaccharides (1 mark)

1 mark

Question 5 (10 marks)

A popular Australian take away meal is "Fish n' chips". The oil used for the deep frying of the food contains the fatty acids: palmitic acid, stearic acid and linoleic acid.

a. Name the common functional group for these fatty acids.

Carboxyl

1 mark

b. The following table shows the melting points for these fatty acids.

Fatty acid	Melting point (°C)
palmitic acid	63
stearic acid	70
linoleic acid	-5

i. With reference to their structure and bonding, explain the difference in melting points between palmitic acid and stearic acid.

Stearic and palmitic acids are both saturated fatty acids which have a straight chain like structure that allows them to form close packing thus being a solid at room temperature.

Palmitic acid ($CH3(CH2)_{14}COOH$) have a shorter hydrocarbon chain than stearic acid ($CH3(CH2)_{16}COOH$) therefore, palmitic acid has weaker dispersion forces between its molecules, resulting in a lower melting point

OR

Stearic acid has a longer hydrocarbon chain than palmitic acid and therefore has stronger dispersion forces between its molecules, resulting in a higher melting point

- Structure: **saturation and length of hydrocarbon chain (or molar mass)** (1 mark)
- Bonding: **Comparing dispersion forces** regarding length (1 mark)

2 marks

ii. With reference to their structure and bonding, explain the difference in melting points between linoleic acid and stearic acid.

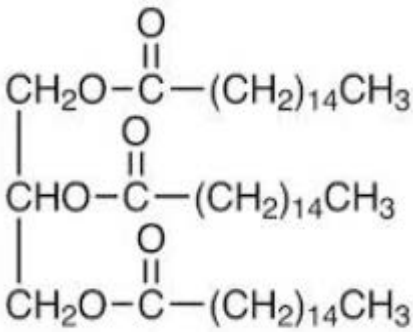
Stearic acid will not form kinks as it is saturated (no C=C double bond), whereas linoleic acid will form kinks at the C=C double bonds due to cis conformation (unsaturated/2 C=C double bonds) in the hydrocarbon chain.

Kinks will prevent close packing of linoleic acid molecules. Therefore, the dispersion forces are weaker between linoleic acid molecules, therefore resulting in a lower melting point than compared to stearic acid.

- Structure: compare **degree of saturation** (presence of C=C only in linoleic acid) and **length** (or number of carbons) (1 mark)
- Bonding: the effect of **kinks on close packing and dispersion forces** (1 mark)

2 marks

c. Another commonly used oil for deep frying is palm oil, which mainly consists of glyceryl tripalmitate. The structural formula is shown below:



i. Name the functional group in the glyceryl tripalmitate molecule.

Ester

1 mark

ii. At the beginning of digestion, glyceryl tripalmitate is broken down into smaller molecules. What is the name given to this general type of chemical reaction?

Hydrolysis /hydrolytic reaction

1 mark

END OF QUESTION AND ANSWER BOOKLET